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Research Article

Understanding encounters between recreationists and livestock guardian dogs as a basis for conflict management

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A B S T R A C T

Farmers in the French Alps increasingly rely on Livestock Guardian Dogs (LGDs) to protect herds in mountain pastures and valleys from a growing wolf population. However, rising sport and recreational activities challenge the coexistence between LGDs and outdoor users. Since 2019, pastoral services in four French departments have conducted surveys – collecting quantitative and qualitative data – to understand recreationists encounters with LGDs.

This research paper examines which factors influence the perception of encounters between recreationists and LGDs. Based on 1587 questionnaires, perceived LGD threats and emotional experiences during and after encounters significantly influenced users' perceptions, both positively and negatively. Awareness of LGDs played a modest mediating role, slightly shifting perceptions towards a good stance, underscoring the importance of awareness campaigns.

Personal experiences also influenced perceptions; first-time encounters and professional visits led respondents to describe encounters as "neutral" rather than "bad", while older users described them as "neutral" rather than "good". Gender and age also played a role: women reported a higher number of "good" encounters. Surprisingly, respondents crossing flocks described encounters as "good" rather than "neutral", challenging recommendations against flock crossing and highlighting adaptation needs.

A key tension lies in contrasting views of mountain areas as non-excludable common land versus excludable resources. Recreationists often see these landscapes as open-access, yet flocks and LGDs introduce barriers, creating exclusion. This reflects three themes: legitimacy (privatisation perceptions), responsibility (accountability conflicts), and socio-ecological issues within herding, recreation, and predator management systems.

Management implications

- Strengthen awareness campaigns led by pastoral services, local authorities, tourist offices, environmental educators, mediators, and eco-guards, as implemented in several French municipalities and protected areas, in order to reduce the pressure on livestock farmers and shepherds.
- Ensure that these awareness campaigns take into account the diversity of pastoral practitioners as well as the variety of contexts in which encounters take place.
- Promote mutual understanding and dialogue between pastoral actors and recreationists.

1. Introduction

Over the past few decades, the recovery of large carnivores (LCs) such as wolves (*Canis lupus*) across Europe has sparked renewed debates about the coexistence of humans and wildlife (Chapron et al., 2014; Pooley et al., 2017). While supportive conservation policies have facilitated wolf recolonisation in regions like the French Alps, this expansion has reactivated and intensified social conflicts, particularly in areas characterised by extensive pastoral systems (Dorresteijn et al., 2015; Redpath et al., 2015). Pastoralism – encompassing sheep, goat and cattle grazing – plays a pivotal role in maintaining High Nature Value (HNV) farmland, sustaining biodiversity, and preserving cultural landscapes (Beilin et al., 2014; Plieninger et al., 2016). However, the same landscapes that benefit from grazing practices also harbour concerns related to livestock depredation and economic vulnerability tied to the presence of predators (Eklund et al., 2017). Consequently, livestock protection measures – most notably Livestock Guardian Dogs (LGDs) – have gained traction across Europe and beyond as a means to reduce depredation and support coexistence (Gehring et al., 2010; Landry et al., 2020).

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However, these interventions can introduce new social frictions. LGDs, typically large and assertive dogs bred or trained to protect herds, not only confront wolves but also interact with people crossing the same landscapes. LGDs generally enjoy more public support than lethal control measures (Mosley et al., 2020) and have been shown to reduce depredation in various contexts (Andelt, 1992; Gehring et al., 2010; Urbigkit & Urbigkit, 2010; VerCauteren et al., 2011). However, in mountainous regions such as the French Alps – where tourism and recreation are economically significant – encounters between LGDs and hikers, mountain bikers and other recreationists can create novel forms of conflict (Landry, 1999; Lescureux & Linnell, 2014). While many perceive LGDs as a critical strategy for limiting wolf attacks on livestock, some visitors experience fear or irritation when confronted by protective dogs. Fear of dogs is quite common in the general population, and being female, having previously been threatened by a dog and being older are all strong predictors of dog-related fear (Boyd et al., 2004). The emotion of "fear" can be felt in the body and in relationships with others and spaces (Tuan, 2013). Fear during encounters can lead to negative attitudes towards pastoral activities and carnivore conservation (Young et al., 2015). Existing research suggests that maintaining spatial buffers between livestock and recreational areas can help reduce such incidents (Mosley et al., 2020). However, those recommendations often originate from large-scale grazing contexts (e.g. in the western United States) and may not translate easily to more spatially constrained European settings. In a context where recreationists are increasing in Europe (Melo et al., 2020), understanding these social dimensions is vital because successful LC management depends not only on technical solutions (e.g., dog training, fencing) but also on public acceptance, perceptions of fairness, and the willingness of different user groups – such as farmers and recreationists – to share a landscape with wildlife (Pooley et al., 2017). A study conducted by Ribert et al. (2024) shows that despite reduced territories and attempts to share mountain spaces, a fragmentation of these areas occurs. More or less fixed and restrictive boundaries (vegetative cover, fences, modifications to trails, etc.) are established and used by shepherds to separate their movements from highly frequented areas (Ribert et al., 2024). The conclusions of this research indicate that recreationists and pastoralism struggle to coexist since the reintroduction of livestock protection dogs in the mountains.

In France, there are currently around 5000 LGDs, reflecting their official status as one of three major herd protection measures – alongside the presence of a shepherd and nighttime enclosures – mandated for eligibility for government compensation (DREAL, 2020). The relationship between shepherds and flocks has now expanded to encompass living with wolves and LGDs, as well as navigating increased interactions with recreationists. This complexity is heightened by the fact that, in the event of an LGD attack on recreationists, legal responsibility may fall on dog owners; following an evaluation, the owner may be asked to euthanise their dog and pay a fine (Mounet, 2007). However, it is often unclear whether the LGD's behaviour or the behaviour of recreationists – or both – drives these encounters. Likewise, it remains difficult to ascertain what determines whether a recreational user perceives the encounter as good or bad, be it the dog's behaviour or individual factors such as prior experience with dogs and fearfulness.

In this context, various organisations representing pastoral interests collaborated on a survey to assess interactions between recreationists and LGDs in the French Alps. We analysed the data from this survey using a mixed-methods approach (quantitative and qualitative) to identify the factors that influence the quality of encounters between LGDs and recreationists, including socio-demographic variables, knowledge about LGDs and pastoralism in general, LGD behaviour, and shepherds' behaviour.

This research paper examines which factors influence the perception of encounters between recreationists and LGDs. this study makes theoretical contributions to people-nature studies and offers practical guidance for encouraging meaningful coexistence between pastoralists, local communities, recreationists and large carnivores.

2. Study area and methods

"My Experience with Livestock Guardian Dogs" was a quantitative and qualitative survey carried out between 2019 and 2021 by the pastoral service team of four French Alpine departments followed by collaborative work with researchers in social sciences.

2.1. Instrument

The survey instrument (Supplementary material A) was divided into six sections to examine various facets of people's encounters with Livestock Guardian Dogs (LGDs): 1. the encounter; 2. the person's reaction, including their emotional and behavioural reaction; 3. the LGD's behaviour towards the people encountered; 4. the context of the encounter, including the weather, the presence or absence of accompanying persons, and the availability of information on the presence or absence of LGDs; 5. information about the respondents, including socio-demographic data, their use of the mountains, and their level of comfort with LGDs; 6. the consequences of the encounter, such as changes in the person's subsequent behaviour, residual emotions, repercussions on their perception of LGDs, etc.

Two open-ended questions – "Can you describe the encounter?" and "Can you tell us more about the encounter?" – allowed respondents to add detail and nuance, although 4.3% of participants chose not to answer them, and some only answered one of them.

2.2. Data collection and sample

The questionnaire was distributed by the pastoral services in four departments of the Auvergne-Rhône-Alpes region (Savoie, Haute-Savoie, Isère and Drôme) between 2019 and 2021. It was available in paper version (notably in tourist offices) and in digital form accessible via a QR code. The pastoral services, as well as tourism professionals and public stakeholders, published the survey via social networks and the media. In the field, nature educators approached all hikers they encountered to invite them to complete the survey, although this method remained limited due to insufficient human resources. Since this was a non-interventional study and participants' anonymity was protected, no ethical approval was required under national regulations. However, consent was obtained from all participants before the survey was administered, and the data were anonymised such that no individuals could be identified. In total, 1603 questionnaires were returned, and 16 surveys (1.2%) were excluded, as they described encounters occurring when LGDs were indoors rather than in a typical outdoor grazing context. The final dataset therefore included 1587 participants.

2.3. Data analysis

a. Multinomial logistic regression

A multinomial logistic regression model was used with R software to determine the factors that influenced the way individuals perceived their encounters with LGDs. The dependent variable in this model was the respondent's evaluation of the encounter, divided into three categories: "good", "neutral" (neither good nor bad in the questionnaire) and "bad".

In order to simplify the logistic regression analysis, groupings were made according to certain criteria associated with specific questions (see supplementary material B entitled "Transformation of variables").

In the absence of a theoretical model to guide the choice of explanatory variables to be included, statistical methods for automatic variable selection were used. The independent variables were selected in two stages. First, a pre-selection was made using bivariate analyses. The independent variables which showed a statistically significant relationship with the dependent variable (the individual's evaluation of the

encounter) were retained as potentially explanatory. A bidirectional stepwise regression approach based on the Akaike Information Criterion (AIC) was then used to identify the final variables to be included in the model. The aim of this method was to select the variables that made a significant contribution to explaining the respondents' perceptions of the encounter. To validate the model, we used McFadden's R^2 to measure its explanatory power and the VIF to identify any problems of multicollinearity between the explanatory variables. The Hosmer-Lemeshow test was also performed to check the goodness of fit of the model (Hosmer et al., 2013).

b. Analysis of open-ended questions

We analysed 2132 responses to two open-ended questions – 1305 responses (82.2% of the sample) to the open-ended item *"Can you describe the encounter?"* and 827 responses (52.1% of the sample) to the open-ended item *"Can you tell us more about the encounter?"* – using NVIVO software, which allows for the systematic processing of large quantities of qualitative data.

The coding produced included four themes in four main categories: (1) information about the encounter; (2) information about the shepherd; (3) information about the LGD; and (4) general considerations (see supplementary material F entitled "Code book"). The first three categories mentioned correspond to topics already addressed in the quantitative section of the survey, providing illustrated examples of how a respondent's attitude might shift during an encounter or how a shepherd's behaviour could influence the respondent's judgment. In contrast, the fourth category allowed for an inductive approach, capturing themes that were not covered by the survey questions. This fourth category was thus divided into six sub-codes: (4.1) the respondent's views on the legitimacy of pastoralism and their own activity, (4.2) responsibility concerns in the event of an LGD-related incident, (4.3) broader issues surrounding pastoralism and LGDs, (4.4) long-term changes in recreationists' attitudes post-encounter, (4.5) prior knowledge about wolves, and (4.6) perceptions of other activities and recreationists.

3. Results

3.1. Quantitative results

Out of the 1587 surveys analysed, the sample consisted of 58% men and 42% women, with a median age of 47 years. Most respondents reported prior experience in mountain environments, with 32% identifying as amateurs, 61% as experienced users, and 6.9% as professionals. Additional socio-demographic characteristics are presented in Appendix E.

Regarding perceptions of encounters with livestock protection dogs (LGDs), there was a predominance of negative perceptions, expressed as "bad" encounters by more than half of the participants (57.3%), followed by "good" encounters (21.5%) and "neutral" encounters (21.2%).

In total, we found that 8 variables made a significant contribution to explaining respondents' perceptions of encounters with LGDs, with 3 variables being detailed by 11 sub-variables (Table 1). Table 1 shows the distribution of these variables among participants (expressed as a percentage for qualitative variables and as an average for age, which is a numerical variable) and the results of the logistic regression.

Our results suggest that the factors that determine the evaluation of the encounter with the LGDs are related to the respondents' socio-demographic characteristics, their individual behaviour, the context of the encounter, the feelings during and after the encounter and the LGDs' behaviour. In contrast, the behaviour of the shepherd did not seem to be a determining factor, but it should also be noted that the shepherd was present in only 29% of the encounters.

A positive coefficient indicates that an increase in the explanatory

Table 1
Logistic regression results: Factors influencing the perception of the encounter.

Multinomial logistic regression results						
Dependent Variable (DV): Perception of the encounter with LGDs Number of criteria: 3						
Parameters	Good vs. Neutral (neither good nor bad)			Bad vs. Neutral (neither good nor bad)		
	Coefficient	Odds Ratio	P-value	Coefficient	Odds Ratio	P-value
Intercept	1.30	3.68	0.0294 **	−1.68	0.19	0.0139 **
Gender [Reference = Women]						
Men (59%)	−0.62	0.54	0.0080 ***	0.01	1.01	0.9680
Age (average 47 years)	−0.02	0.98	0.0194 **	0.01	1.01	0.1536
Passing through the flock (15%)	0.60	1.82	0.0383 **	−0.22	0.80	0.5067
Informed of the presence of LGD(s) (47%)	0.79	2.2	0.0004 ***	0.46	1.58	0.0539 *
First encounter with an LGD (17%)	−0.22	0.80	0.4459	−1.01	0.37	0.0005 ***
Visiting the mountains for professional reasons (13%)	−0.1	0.91	0.7629	−0.72	0.49	0.0384 **
Feelings during the encounter[a]						
Calm (13%)	1.64	5.14	0.0000 ***	−1.31	0.27	0.0360 **
Relaxation (6%)	1.09	2.96	0.0103 **	−0.35	0.71	0.7031
Distraction (3%)	1.6	4.99	0.0035 ***	−0.77	0.46	0.5294
Distrust (32%)	−0.43	0.65	0.0659 *	−0.47	0.62	0.0580 *
Fear/Panic/Terror (49%)	−1.29	0.28	0.0005 ***	1.45	4.28	0.0000 ***
Feelings after the encounter[a]						
Incomprehension (29%)	−1.58	0.21	0.0138 **	0.72	2.05	0.0154 **
Anger (43%)	−2.19	0.11	0.0080 ***	1.77	5.87	0.0000 ***
Indifference (8%)	−0.61	0.54	0.0935 *	−1.35	0.26	0.0198 **
Understanding (32%)	0.3	1.35	0.3559	−1.72	0.18	0.0000 ***
LGD's behaviour[a]						
Barking, growling (74%)	−0.65	0.52	0.0053 ***	0.28	1.33	0.3792
Being or becoming threatening (44%)	−1.54	0.21	0.0001 ***	1.9	6.67	0.0000 ***

***: Significant at 1% **: Significant at 5% *: Significant at 10%.
The logistic regression model explained 57% (McFadden's R2) of the variance in the perception of encounters with LGDs. An evaluation of the goodness fit of the specified model indicates that it is significant overall (likelihood ratio test (null model VS estimated model): $Chi^2 = 1389.212$; ddl = 34; p-value <0.001) and well-calibrated (Hosmer-Lemeshow: $Chi^2 = 10.3418$; ddl = 16; p-value = 0.8482). An examination of the variance inflation factors (VIF) of the explanatory variables shows that our model does not present a multicollinearity problem (see supplementary material D entitled "Multicollinearity test").
[a] More than one answer possible.

variable is associated with an increase in the chances of an encounter being perceived as "good" or "bad" rather than "neutral". The odds ratio, on the other hand, quantifies the extent to which an increase or the presence of a predictor variable influences the ratio of the chances that an encounter will be perceived as "good" or "bad" rather than "neutral".

In terms of socio-demographic characteristics, the results show that women were 1.9 times more likely than men to rate encounters as "good" rather than "neutral". In addition, it appears that the older the respondents were, the less likely they were to describe the event as "good" rather than "neutral".

With regard to the respondents' behaviour, passing through the flock increased the probability of rating the encounter as "good" by 1.8 times, rather than as "neutral".

Regarding the context of the encounter, being informed of the presence of the LGDs increased the odds of rating the encounter as "good" by 2.2 times, rather than rating it as "neutral". A first encounter with LGDs, and the fact that the respondent was visiting the mountains for professional reasons, seemed to have led more people to consider the encounter as "neutral" rather than "bad".

Positive emotions experienced during the encounter, especially feelings of relaxation and distraction, seem to have increased the likelihood of the encounter being perceived as a positive experience rather than neutral.

Ultimately, the negative feelings during and after the encounter, as well as the threatening behaviour of the LGDs, were the determining factors for both positive and negative perceptions compared to neutral ones. Both negative feelings and the behaviour of the LGDs increased the likelihood of rating the encounter as "bad" rather than "neutral" and decreased the likelihood of rating it as "good" rather than "neutral". In other words, these elements were significant in contributing to a negative evaluation of encounters with LGDs. In terms of feelings of fear or panic during the encounter, the odds ratio between "good" and "neutral" is 0.28, which means that this increased the probability of describing the encounter as "neutral" rather than "good" by 3.57 (1/0.28). This feeling also increases the chance of describing the encounter as "bad" rather than "neutral" by 4.28. In this case, we can deduce that those who felt panic or fear were 15.29 (3.57 * 4.28) times more likely to rate the encounter as "bad" rather than "good" (the method used to calculate this is explained in the supplementary material C). Similarly, feelings of incomprehension and anger after the encounter increased the odds of describing the encounter as "bad" rather than "good" by 9.76 and 53.36 times respectively. Lastly, the LGD's threatening behaviour increased the likelihood of describing the encounter as "bad" rather than "good" by a factor of 31.75.

3.2. Qualitative results

A core tension emerging from our data lies in the contrasting perceptions of mountain areas as a non-excludable versus excludable resource. Many recreationists view these pastoral landscapes as a common good, open to everyone. At the same time, the presence of livestock and Livestock Guardian Dogs can effectively restrict access or make some visitors feel unwelcome, introducing a sense of excludability. Beneath the various interpretations of mountain spaces as common goods, we will focus our analysis on three sub-codes that introduce duality between non-excludable versus excludable resources: "Questioning legitimacy" shows how the presence of flocks and LGDs can be viewed as effectively privatising what many see as a non-excludable landscape, "Who is responsible?" reveals conflicting views on shepherds' accountability versus visitors' duties in a nominally public domain that becomes partially excludable, and "Understanding wider issues" situates these conflicts within broader socio-ecological systems, highlighting how decisions about herding, recreation, and predator management further recalibrate the boundaries between open-access and restricted areas.

a. Questioning legitimacy

Based on the qualitative answers, it is clear that recreationists question the legitimacy of pastoralism or their own activity in the mountains. Many respondents view the mountains as a non-excludable space, freely available to all: "*For me, the mountains are a place of freedom and mutual respect for all users and activities*". For some, however, pastoralism – especially when flocks roam with LGDs – excludes other users and feels more excludable in practice: "*Flocks roaming freely in the wilderness can lead to a kind of privatisation of certain areas because they are so closely guarded."* Here, the "freedom of some" implies "the exclusion of others", revealing how a commonly perceived open-access area can suddenly feel privatised by grazing practices and LGDs.

In this context, many respondents do not consider LGDs themselves to have legitimacy. The fact that hikers' companion dogs are usually restricted bolsters their perception that pastoralists' dogs are given preferential treatment. The absence of reciprocity between shepherds and other mountain users calls into question what is legitimate for shepherds and their animals: "*[Shepherds] should be bound by the same kind of obligations towards their dogs as we are, within the limits of their professional activity. For example, dogs should not be roaming out of sight or control of the shepherd. They should call the dog back […]."*

Moreover, some recreationists s highlighted safety concerns, questioning whether it is acceptable for LGDs to pose a risk: "*Humans [should not have to] fear the presence of dogs"* or "*It is unacceptable to frighten hikers and disturb the peace of the mountains with wild animals barking*!" This prompts us to highlight the issue of safety, which came up regularly in the respondents' comments. Some respondents questioned their own safety or the safety of vulnerable people, such as children, on the mountain pastures in the presence of LGDs. References, like the one below, to a possible future incident involving children can be used to cast doubt on the legitimacy of the presence of LGDs and their use by shepherds. "*The total indifference on his part, imagining that I could have been with my children (who could have had a reaction leading to an attack without me being able to do anything), makes me feel immense anger towards him."*

Ownership considerations also shape legitimacy. Ownership can be seen as a fact established by law: "*[…] as tourists, we are on the shepherds' land*" or a feeling that there is a territory or area dedicated to a particular activity. Sometimes it is the shepherds who have ownership: *"They're doing their job, it's up to us to respect the area they work in*" and "*The mountains belong to the animals*". At other times, recreationists have ownership: "*It's unacceptable that the sheep enclosure is in the middle of the path and that the dogs are in the way*". This last quote reflects the feeling shared by a number of respondents that the path – especially if it is signposted – is a place where recreational activities have precedence. Consequently, it is less legitimate for the flock to be there, as expressed by this user: *"[…] I don't understand why the dogs and flocks are on very busy paths where it's impossible to get around them because of the terrain"*. Some respondents argue for the continued non-excludable access of these trails: *"Something must be done to guarantee freedom of movement on the GR footpaths*[1] *!*" The sense that flocks and LGDs occupy "all the areas" leads a few people to foresee the end of certain leisure activities: "*Where can we go without worrying about encountering these dogs? Because all the areas seem to belong to them …*"

Finally, since sharing the mountains seems to have become impossible or near impossible for some recreationists, some respondents suggested a discontinuation of pastoralism: "*I'm now restricted to where I can go, and I can't help but resent the profession. I don't eat lamb anymore!*" or of the presence of LGDs: "*I hope that this practice of using mountain dogs will be stopped forever […]*" or a discontinuation of leisure activities in the

[1] GR footpaths refer to a network of long-distance walking trails in France. GR is an abbreviation of 'Chemins de Grande Randonnée' in French, translating to 'Great Hiking Trails' in English.

area: "*Hikers and mountain bikers should think about going elsewhere*." Given that grazing is taking over the area and making it inaccessible, some people foresee the end of leisure activities: "*As a mountain professional, I'm worried about the impact of dogs on tourism …"* and "*In such cases, it's obvious that we're in the way and that, as hikers, we shouldn't be in the mountains. It's a real shame, because us spending time here also enables mountain people to make a living from tourism*". Overall, this tension reflects a clash between perceiving these landscapes as non-excludable common land (free for everyone) and the practical reality that livestock grazing – along with LGDs – may effectively exclude or deter certain user groups.

b. Who is responsible?

When an encounter with LGDs is perceived as problematic, respondents often attribute responsibility to specific stakeholders. Responsibility was sometimes attributed to the shepherds and/or livestock farmers, whose roles were often not differentiated by the respondents. In this instance, the owner's responsibility towards their dog is at the heart of the issue: *"[…] to be honest, it's a real problem because unfortunately a dog is still a dog, it can be unpredictable and even if it's trained you have to keep an eye on it […]! It's logical that a dog needs an owner!!!!*"

The shepherd is therefore considered to be the LGD's rightful owner and must be present and active during encounters between LGDs and recreationists. *"In the same way that dogs should be kept on a leash, a guardian dog should only be permitted if the shepherd is there."* Yet respondents regularly highlighted the shepherd's absence: "*The shepherd should be there"; "I didn't see a shepherd. Was there one?*"

If the shepherd was present, according to the various respondents, they were expected to act: *"[…] it's their responsibility to control their animals. […]*" More specifically, they were supposed to adopt the correct attitude described here: "*There are also many KIND shepherds who call their dogs back when walkers approach*". And if they do not do so, they could be held responsible for the incident: "*A shepherd was nearby and didn't intervene. I didn't understand why he didn't [react]. This is deliberate endangerment of others*". However, to nuance those points of view, a small minority of recreationists explained that the shepherd has no direct authority over the LGD: "*I have heard that they [the shepherds] meet their dogs just before they go up to the mountain pastures. I can understand why accidents happen*".

Apart from the encounter itself, shepherds and livestock farmers were considered to be responsible for LGD training. Respondents therefore had an opinion on how the LGDs should be trained: "*These dogs are left half wild and don't know the difference between the potential threat of a wolf and humans*". While in some cases the LGDs' training was considered to be poorly conducted or incomplete, some respondents accused the shepherds/livestock farmers of breeding dangerous LGDs in order to discourage people from visiting the mountains. As one recreational user put it: "*Some don't hesitate to keep dogs that are aggressive towards people, 'that way they leave us alone', they say.*" One hiker stated: "*The bad encounter was meeting this shepherd who trains his dogs to chase hikers."*

Responsibility can also be attributed to decision-making bodies. Respondents sometimes referred to a specific authority, ranging from the government to the tourist office, or even the town council. Some respondents talked about a "system" or a "they" that can be ascribed to these bodies. "*The measures taken by the government are inadequate, and some are even irresponsible"*, said one recreational user.

Lastly, some recognise that the shepherd or livestock farmer was responsible, or at least not solely responsible, and proposed that responsibility should be shared: *"[…] everyone must be responsible: not only hikers, tourists, cyclists, etc., by respecting the areas and those who occupy them, but also the livestock farmers/shepherds by controlling their dogs better.*" Alternatively, recreationists were sometimes held responsible, as this respondent explains: "*We have to know and respect the places we pass through. We must therefore be aware of the potential "risks" and not disturb or upset the flocks and shepherds (who are working!!)*". In this instance, it is up to those passing through the area to pay attention to the herds and their LGDs: "*It's up to us to share their territory discreetly*". This attitude is nevertheless a minority among respondents.

This highlights a tension in perceived accountability, illuminating how different stakeholders define (and sometimes shift) responsibility for ensuring safe coexistence on these landscapes, which are perceived as partially excludable by respondents.

c. Understanding wider issues

The respondents' capacity to situate their encounters in broader contexts also appeared to influence their overall evaluation. For example, in their comments, some respondents acknowledged the merits of pastoralism and the usefulness of livestock farming as a service for the ecosystem, local farming, and as a feature of the landscape and traditional way of life: "[we] *STAND BY the shepherds who graze their flocks to maintain our beautiful natural environment*". Sometimes, respondents even saw how this complemented their activities, as explained here: "*We need to teach visitors that nature does not belong to them and that these flocks 'shape the landscape', grazing the grass and clearing areas for the benefit of us all"*.

Respondents who talked about wider issues, also began by mentioning the current context regarding predation: "*None of this would exist if the wolf had not been allowed to thrive, or even reintroduced to begin with!*". Fewer respondents acknowledged the role of the LGDs by linking it to predators: "*Mountain dogs are one of the solutions for helping herds coexist with wolves in mountain pastures"*. Lastly, some respondents mentioned the increasing number of issues faced by shepherds, for instance, predators, current economic problems, encounters with hikers and other recreationists. "*I fully understand that the presence of wolves has changed the situation and that it's legitimate for shepherds to protect their flocks, their animals and their work"*, explained one respondent. "*How do you ensure that sheep can graze on mountain grass without getting their throats slit?*", asked another.

Overall, by recognising the intersection of non-excludable recreational spaces and essential pastoral operations, a small minority of respondents see LGDs as part of a broader strategy for the coexistence of humans and wildlife. However, many continue to struggle with what they perceive as an excludable barrier to their own activities, underscoring how deeply these encounters influence people's feelings of inclusion and safety.

4. Discussion and conclusion

In this research, we combined quantitative and qualitative approaches to better understand the determinants of positive, neutral, or negative perceptions of encounters between recreationists and Livestock Guardian Dogs (LGDs) in the French Alps. As encounters between LGDs and recreationists are likely to increase, a clearer understanding of these interactions in pastoral areas may help identify both barriers and leverage points for coexistence. As there is still limited literature on the topic of CPT, we draw parallels with carnivore management literature, where conflicts are similarly shaped by fear, anger, and divergent stakeholder interests (Carter et al., 2012; Pooley et al., 2017).

4.1. LGD behaviour and emotional responses

In examining the quantitative results of the study, we see that the dynamics of the encounter itself play a central role in shaping people's reactions. Unsurprisingly, our findings reveal that LGD behaviour has the greatest impact on how the encounter is perceived: a more threatening LGD is associated with a worse perception, whereas a more passive one contributes to a better perception. However, this relationship seems moderated by whether respondents were aware of the LGD's presence, as those who anticipated encountering a dog reported better

(rather than neutral) experiences. Such prior knowledge allows individuals to adapt their behaviour – by changing their route, preparing for a potential encounter, or even turning back – reducing surprise and fostering acceptance. This result aligns with research on large carnivores, where risk perception and prior knowledge significantly affect attitudes as shown by Carter et al. (2012).

The emotions felt during and after the encounter also play a crucial role in shaping both positive and negative perceptions. Fear or panic during the encounter increased the likelihood of describing it as "bad" whereas anger afterwards raised "bad" ratings even more dramatically. Negative encounters with LGDs – much like those with predators – can trigger intense emotional reactions, including fear, anger, and incomprehension, which can shape attitudes toward entire species or conflict-mitigation strategies (Vaske & Roemer, 2013). If left unaddressed, these emotions can escalate conflicts among people; in carnivore studies, anger or frustration is often redirected towards wildlife (Peterson et al., 2010). Here, by contrast, such emotions may be directed at shepherds, authorities, or other perceived responsible parties. Swiftly addressing anger is therefore essential to avoid fuelling tension between these groups.

Prior experience with LGDs.Our results also suggest that experience significantly shapes attitudes. Prior exposure to LGDs and familiarity with pastoral landscapes appear to influence perceptions, much like prior experience influences tolerance towards carnivores in human-wildlife studies. Experience is a common focus in studies assessing tolerance towards carnivores; for instance, personal experience and length of residence in an area both strongly influence attitudes (Kansky et al., 2014). In our case, a more positive attitude was observed among those visiting the mountains for professional reasons, potentially because they perceive the mountains as a working environment where pastoral activities, including LGDs, coexist. This is not the case for recreationists because "recreational perceptions of mountains are prevailing over a pastoral mindset that has become increasingly misunderstood and invisible" (Ribert et al., 2024). Recreationists often view the mountains as a boundless "elsewhere" (Debardieux, 2001), whereas those working in the mountains recognise ownership and obligations (Dalla Bernardina, 2003), thereby normalising the presence of LGDs and accepting their diverse behaviours. More surprisingly, first-time encounters with LGDs were more frequently rated as "neutral" rather than "bad", likely because participants lacked prior experiences to compare, making it difficult to evaluate the quality of encounters.

4.2. Socio-demographic variables

Gender and age also played a role. W omen in our sample exhibited more positive attitudes toward LGDs, contradicting some studies suggesting that women typically report higher levels of fear or discomfort around animals, and especially dogs (Boyd et al., 2004)– possibly due to socialisation, personal experience, or both. However, such results can vary by cultural context and the specifics of each encounter (Herzog, 2007; Johansson & Frank, 2011). Finally, older recreationists tended to describe their experiences as neutral rather than positive. B Boyd et al. (2004) demonstrate that older people generally have a greater fear of dogs. Furthermore, many of the individuals in our study knew the mountains before LGDs were reintroduced, and their long-standing familiarity with the area may make them less willing to accept the changes brought about by LGDs.

4.3. legitimacy

An important qualitative finding showed that mountain pastures occupy a contradictory space in the public imagination: they are not only widely viewed as open-access (i.e., non-excludable) but are also legally or customarily managed for pastoral use, rendering them partly excludable (Eychenne & Lazaro, 2014; Ribot & Peluso, 2003). Many recreationists regard the mountains as "a place of freedom", believing that they "belong to everyone" (Dalla Bernardina, 2003), whereas pastoralists (who may own or lease the land) see a workplace needing protection from predators (Barrière & Bes, 2017; Baumont, 2004). Here, the presence of LGDs can be interpreted as "privatising" or fencing off common land in ways that limit freedom of movement – effectively transforming a nominally non-excludable good into something more excludable (Harribey, 2011; Ostrom & avecBaechler, 2010). Meanwhile, from the pastoralist perspective, the presence of dogs is a legitimate response to wolf predation, supported by both policy and local customs (Landry et al., 2020). This clash of interpretations about "who owns the space" underlies many tensions, echoing fundamental conflicts over common-pool resources (Mormont, 2006; Ostrom & avecBaechler, 2010), and shedding light on deeper questions of legitimacy in pastoral spaces.

In processes of legitimisation, different stakeholders often rely on their own standards of what is "fair", using these to validate their actions and to contest those of others (Garin & Richard-Ferroudji, 2008). The conflict of legitimacy thus arises from asymmetries in knowledge (e.g., legal rights, ecosystem services) and from differing understandings of whether pastoral activities belong in a space romantically perceived as wild (Descola, 2004; Hodder & Bullock, 2009). Shepherds, for example, generally know the regulatory framework governing mountain lands, whereas many recreationists do not. In France, livestock farmers typically own their land or use it under formal or informal agreements (Barrière & Bes, 2017). Regular activity on these lands can foster a sense of comfort and ownership in shepherds (Baumont, 2004), but many recreational visitors remain unaware of this and interpret the behaviour of certain shepherds as an undue claim on space that contradicts their vision of the mountains. Some individuals opposed the notion that LGDs should roam freely, associating certain canine behaviours with aggression and arguing that dogs must stay under the shepherd's direct control, even though LGDs are permitted to operate at a distance from their handler and sometimes need to leave the herd to ward off disruptions (Bobbé, 1999; Landry et al., 2020). However, in opposition, some respondents did grasp the broader issues surrounding LGDs and recognised the benefits of pastoralism on the local ecosystem and the need for LGDs, thereby showing empathy for pastoral practices.

4.4. Practical implications

Addressing this diversity of experiences with LGDs and pastoral areas thus demands a nuanced approach, rather than a one-size-fits-all strategy, to meet the needs of all recreationists.

Our findings indicate that providing warnings about LGDs and clarifying behavioural norms – such as whether to pass through a flock or take a detour, or how to interpret LGD behaviour – can help reduce hostility and perceived exclusion (Beilin et al., 2014; Eklund et al., 2017). It is equally important to foster empathy for the emotions recreationists experience after an encounter with LGDs, as well as to cultivate empathy among these users towards pastoral communities more broadly – an approach highlighted in human-wildlife conflict research emphasising the role of empathy in effective carnivore conservation (Kansky & Kidd, 2024).

Providing information about why LGDs are present and how to respond when encountering them emerged as crucial for improving visitor experiences. Participants who were aware of the presence of LGDs in advance tended to rate their encounters as "good" rather than "neutral". However, only 47% of respondents reported knowing that LGDs would be present, indicating a significant need to improve awareness. Future awareness campaigns must consider the diverse profiles of recreationists, particularly considering their experiences with LGDs, as this significantly influences participants' perceptions of encounters in complex ways. Furthermore, recommendations in LGD-awareness campaigns are not always straightforward: for instance, our data suggests that respondents who passed through the flock reported better experiences (ranked as "good" rather than "neutral"),

contradicting many LGD-awareness recommendations against flock crossing. This outcome may stem from scenarios where the dogs did not behave as deterrents – e.g. barking or running – and instead kept a distance from the flock. Recreationists who are not obliged to take a long detour might thus view the encounter more favourably, as it causes minimal disruption to their route. However, this finding is particularly intriguing within the ongoing debate on best practices for herd crossing. In some high-traffic mountain pastures, LGDs appear more accustomed to hikers who remain on the designated trail and can become more alarmed if people deviate from it to circumvent the herd. Furthermore, many hikers do not move far enough to leave the LGD's safety zone, rendering their detour ineffective for lowering the dogs' stress (Pastoral Services). Finally, awareness-raising campaigns targeting recreationists should include information about the specific role and status of LGDs, which could help them better understand the system involving the shepherd, their flock, wolf management, and LGDs. This aligns with the need for awareness campaigns to be adaptive or context-specific, similarly to strategies advocated in large carnivore management, where a single set of universal guidelines often fails to capture variations in local dog training, terrain, or social norms (Eklund et al., 2017; Kansky et al., 2014). This campaign should be led by pastoral services, municipalities and tourist offices, as well as nature educators, nature mediators and eco-guards (municipal employees), as has been done in several French municipalities and protected areas, to alleviate the burden on breeders and shepherds.

Equally important is fostering two-way communication and mutual empathy (Perrin-Malterre et al., 2021, pp. 375–393). In particular, pastoral communities have established systems for reporting incidents (nipping or biting) to local authorities (Direction Départementale des Territoires) or pastoral associations – 44 incidents were documented in 2022 alone (Réseau Pastoral Auvergne-Rhône-Alpes, 2022). Such systems can help address fear and anger more promptly by providing factual information on LGD use and offering avenues for conflict resolution. As demonstrated in the municipality of Revel, in the Isère department, it is also possible to establish spaces for dialogue between the various stakeholders who interact in mountain areas. The municipality has implemented two annual meetings: one for preparing the summer grazing season and another for reviewing it. These meetings bring together stakeholders from pastoralism, hunting, refuge warden, trail running, and hiking activities. This initiative exemplifies how dialogue and collective problem-solving can foster empathy, clarify the role of LGDs in protecting flocks, and ultimately improve the overall acceptance of coexistence strategies.

5. Conclusion

In this study, we combined quantitative and qualitative analyses to explore the complex dynamics of encounters between recreationists and Livestock Guardian Dogs (LGDs) in the French Alps. Our results show that most encounters were perceived negatively by recreationists, but this finding may be influenced by selection (Hernán et al., 2004) since individuals with negative experiences could be more motivated to participate. In fact, the survey was designed by pastoral services partly to document troublesome incidents – like LGD bites – and to implement mediation – between recreationists and shepherds – where needed. Moreover, recall bias (Baumeister et al., 2001) may also have played a role, given people's tendency to emphasise and remember negative events more vividly than neutral or positive ones. These two biases likely contributed to the high proportion of negative perceptions. Another limit is categorization of LGDs as “threatening” because it is particularly subjective. This label does not specify the exact behaviors deemed threatening, which could range from a dog merely approaching a respondent, to pushing or nipping, or simply barking. Apart from actual bites, other behaviors may be considered acceptable in LGDs, given their role in deterring disturbances to the herd (Bobbé, 1999).

Despite these limits, our research indicates that while the central aspect in evaluating encounters with LGDs remains the perceived behaviour of the LGDs, there are many other factors influencing the perception of recreationists, such as emotional responses, experience, prior awareness of the presence of LDGs, underscoring the importance of tailored information campaigns and context-specific behavioural guidelines. At the same time, legitimacy conflicts arise from divergent views of pastoral lands as common (non-excludable) resources versus working environments that require protection measures. Addressing both the emotional dimensions of LGD encounters and the knowledge asymmetries between pastoralists and recreationists is pivotal for fostering coexistence and sharing landscapes. By emphasising adaptive awareness strategies, empathy-building initiatives, and recognition of the pastoral context, our work offers a pathway to reconcile evolving interactions between recreationists and pastoral communities in multifunctional mountain regions.

The research perspective would be to conduct in-depth interviews to better understand the role of emotions in the perception of encounters with LGD. This would also make it possible to clarify which LGD behaviours should—or should not—be perceived as threats, underscoring the need for continued awareness-raising on appropriate LGD conduct. The same study could also be conducted in other Alpine countries in order to compare results and analyse the importance of LGD acceptability.

CRediT authorship contribution statement

Noémie Bailly: Writing – review & editing, Writing – original draft, Project administration, Formal analysis, Data curation, Conceptualization. **Lou Lecuyer:** Writing – review & editing, Formal analysis, Data curation, Conceptualization. **Moussa Ba:** Writing – review & editing, Formal analysis, Data curation. **Clémence Perrin-Malterre:** Writing – review & editing, Supervision, Formal analysis, Data curation, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Supplementary data

Supplementary data to this article can be found online at https://doi.org/10.1016/j.jort.2026.101043.

Data availability

The data that has been used is confidential.

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